

This document provides additional assistance with wiring your Extron IP Link Pro Control Processor to your device. Different components may require a different wiring scheme than those listed below.

For complete operating instructions, refer to the user's manual for the specific IP Link Pro Control Processor or the documentation supplied by the manufacturer of the controlled device.

For more information on using Global Scripter Modules, refer to the "[Guide to Using Scripter Modules](#)" document.

Device Specifications

Device Type: Display
Manufacturer: LG Electronics
Firmware Version: N/A
Model(s): 55VH7B, 55VH7B-H

Tested on the Following Software and Firmware Versions

IP Link Pro Control Processor Firmware	Global Scripter Version
3.02.0000-b013	2.2.0

Version History

Module Version	Date	Notes
1_0_1_0	10/5/2018	Added model 55VH7B-H.
1_0_0_0	7/27/2017	Initial Version

Module Notes

- Unidirectional variable must be set to 'True' if status is not required. Default value is 'False'.
Example: `InterfaceName.Unidirectional = 'True'`
- connectionCounter variable must be set to the number of queries that will be sent to the device before displaying 'Disconnected' if no response is received. Default value is 15.
Example: `InterfaceName.connectionCounter = 5`
- **Only for serial communication:** DeviceID variable must be set accordingly. Default value is '1'. DeviceID ranges from '1' –' 99' and 'Broadcast'. If DeviceID is set to 'Broadcast', status is unavailable.
Example: `InterfaceName.DeviceID = '1'`
- Ethernet communication does not support status.

Supported Classes and Examples

SerialClass
<code>InterfaceName = ModuleName.SerialClass(ProcessorName, 'COM1', Model='55VH7B')</code>
SerialOverEthernetClass
<code>InterfaceName = ModuleName.SerialOverEthernetClass('192.168.254.254', 2001, Model='55VH7B')</code>
EthernetClass
<code>InterfaceName = ModuleName.EthernetClass('192.168.254.254', 9761, Model='55VH7B')</code>

Control Commands (Serial)

Format with Qualifier:

```
InterfaceName.Set(Command, Value, {'Qualifier Key': 'Qualifier Value'})
```

Format without Qualifier:

```
InterfaceName.Set(Command, Value)
```

Command AspectRatio ¹	Value '4:3'	Value '16:9'	
Qualifier Key 'Device ID'	Qualifier Value '1' – '99'	Qualifier Value 'Broadcast'	
# AspectRatio example InterfaceName.Set('AspectRatio', '4:3', {'Device ID': 'Broadcast'})			
Command AutoImage	Value None		
Qualifier Key 'Device ID'	Qualifier Value '1' – '99'	Qualifier Value 'Broadcast'	
# AutoImage example InterfaceName.Set('AutoImage', None, {'Device ID': 'Broadcast'})			
Command ExecutiveMode	Value 'On'	Value 'Off'	
Qualifier Key 'Device ID'	Qualifier Value '1' – '99'	Qualifier Value 'Broadcast'	
# ExecutiveMode example InterfaceName.Set('ExecutiveMode', 'On', {'Device ID': 'Broadcast'})			
Command Input	Value 'AV' 'DVI-D'	Value 'Component' 'HDMI'	Value 'RGB' 'DisplayPort'
Qualifier Key 'Device ID'	Qualifier Value '1' – '99'	Qualifier Value 'Broadcast'	
# Input example InterfaceName.Set('Input', 'AV', {'Device ID': 'Broadcast'})			
Command Keypad	Value '0' – '9'		
Qualifier Key 'Device ID'	Qualifier Value '1' – '99'	Qualifier Value 'Broadcast'	
# Keypad example InterfaceName.Set('Keypad', '0', {'Device ID': 'Broadcast'})			
Command MenuNavigation	Value 'Up' 'Right'	Value 'Down' 'Enter'	Value 'Left' 'Menu'

	'Back'	'Exit'
Qualifier Key 'Device ID'	Qualifier Value '1' – '99'	Qualifier Value 'Broadcast'
# MenuNavigation example InterfaceName.Set('MenuNavigation', 'Up', {'Device ID': 'Broadcast'})		
Command Power	Value 'On'	Value 'Off'
Qualifier Key 'Device ID'	Qualifier Value '1' – '99'	Qualifier Value 'Broadcast'
# Power example InterfaceName.Set('Power', 'On', {'Device ID': 'Broadcast'})		
Command TileMode	Value None	
Qualifier Key 'Device ID'	Qualifier Value '1' – '99'	Qualifier Value 'Broadcast'
Qualifier Key 'Column'	Qualifier Value '1' – '15'	
Qualifier Key 'Row'	Qualifier Value '1' – '15'	
# TileMode example InterfaceName.Set('TileMode', None, {'Device ID': 'Broadcast', 'Column': '1', 'Row': '1'})		
Command TilePosition ²	Value None	
Qualifier Key 'Device ID'	Qualifier Value '1' – '99'	Qualifier Value 'Broadcast'
Qualifier Key 'Tile ID'	Qualifier Value '1' – '225'	
# TilePosition example InterfaceName.Set('TilePosition', None, {'Device ID': 'Broadcast', 'Tile ID': '1'})		

¹ This command is only available through RGB Input.² Tile ID value must not exceed Row * Column. For a 3*3 grid, the Tile ID should be assigned as follows:

1	2	3
4	5	6
7	8	9

Status Available (Serial)

For all commands, call Update to receive the latest status. ConnectionStatus does not support the Update function and is triggered by the device providing a successful response to other Update function calls.

Format with Qualifier:

```
InterfaceName.Update(Command, {'Qualifier Key': 'Qualifier Value'})  
Value = InterfaceName.ReadStatus(Command, {'Qualifier Key': 'Qualifier Value'})  
InterfaceName.SubscribeStatus(Command, {'Qualifier Key': 'Qualifier Value'}, FeedbackHandler)  
FeedbackHandler will be called only when the specified qualifier gets a new status.
```

Format without Qualifier:

```
InterfaceName.Update(Command)  
Value = InterfaceName.ReadStatus(Command)  
InterfaceName.SubscribeStatus(Command, None, FeedbackHandler)  
FeedbackHandler will be called when any qualifier gets a new status.
```

Command	Value	Value	Value
AspectRatio	'4:3'	'16:9'	
Qualifier Key	Qualifier Value	Qualifier Value	
'Device ID'	'1' – '99'	'Broadcast'	
# AspectRatio examples InterfaceName.Update('AspectRatio', {'Device ID': 'Broadcast'}) Value = InterfaceName.ReadStatus('AspectRatio', {'Device ID': 'Broadcast'}) InterfaceName.SubscribeStatus('AspectRatio', None, FeedbackHandler)			
Command	Value	Value	Value
ExecutiveMode	'On'	'Off'	
Qualifier Key	Qualifier Value	Qualifier Value	
'Device ID'	'1' – '99'	'Broadcast'	
# ExecutiveMode examples InterfaceName.Update('ExecutiveMode', {'Device ID': 'Broadcast'}) Value = InterfaceName.ReadStatus('ExecutiveMode', {'Device ID': 'Broadcast'}) InterfaceName.SubscribeStatus('ExecutiveMode', None, FeedbackHandler)			
Command	Value	Value	Value
Input	'AV'	'Component'	'RGB'
	'DVI-D'	'HDMI'	'DisplayPort'
Qualifier Key	Qualifier Value	Qualifier Value	
'Device ID'	'1' – '99'	'Broadcast'	
# Input examples InterfaceName.Update('Input', {'Device ID': 'Broadcast'}) Value = InterfaceName.ReadStatus('Input', {'Device ID': 'Broadcast'}) InterfaceName.SubscribeStatus('Input', None, FeedbackHandler)			
Command	Value	Value	
Power	'On'	'Off'	
Qualifier Key	Qualifier Value	Qualifier Value	

'Device ID'	'1' – '99'	'Broadcast'
<pre># Power examples InterfaceName.Update('Power', {'Device ID': 'Broadcast'}) Value = InterfaceName.ReadStatus('Power', {'Device ID': 'Broadcast'}) InterfaceName.SubscribeStatus('Power', None, FeedbackHandler)</pre>		

Control Commands (Ethernet)

Format with Qualifier:

```
InterfaceName.Set(Command, Value, {'Qualifier Key': 'Qualifier Value'})
```

Format without Qualifier:

```
InterfaceName.Set(Command, Value)
```

Command AspectRatio ¹	Value '4:3'	Value '16:9'	
# AspectRatio example InterfaceName.Set('AspectRatio', '4:3')			
Command AutoImage	Value None		
# AutoImage example InterfaceName.Set('AutoImage', None)			
Command ExecutiveMode	Value 'On'	Value 'Off'	
# ExecutiveMode example InterfaceName.Set('ExecutiveMode', 'On')			
Command Input	Value 'AV' 'DVI-D'	Value 'Component' 'HDMI'	Value 'RGB' 'DisplayPort'
# Input example InterfaceName.Set('Input', 'AV')			
Command Keypad	Value '0' – '9'		
# Keypad example InterfaceName.Set('Keypad', '0')			
Command MenuNavigation	Value 'Up' 'Right' 'Back'	Value 'Down' 'Enter' 'Exit'	Value 'Left' 'Menu'
# MenuNavigation example InterfaceName.Set('MenuNavigation', 'Up')			
Command PowerOff	Value None		
# PowerOff example InterfaceName.Set('PowerOff', None)			
Command TileMode	Value None		
Qualifier Key	Qualifier Value		

'Column'	'1' – '15'
Qualifier Key 'Row'	Qualifier Value '1' – '15'
# TileMode example InterfaceName.Set('TileMode', None, {'Column': '1', 'Row': '1'})	
Command TilePosition ²	Value None
Qualifier Key 'Tile ID'	Qualifier Value '1' – '225'
# TilePosition example InterfaceName.Set('TilePosition', None, {'Tile ID': '1'})	

¹ This command is only available through RGB Input.

² Tile ID value must not exceed Row * Column. For a 3*3 grid, the Tile ID should be assigned as follows:

1	2	3
4	5	6
7	8	9

Cable and Adapter Requirements

Captive Screw to Female DB9 RS-232 Serial Cable

Notes for the Device

Serial communication

Port Type: RS-232

Baud Rate: 9600

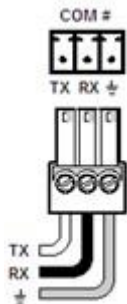
Data Bits: 8

Parity: None

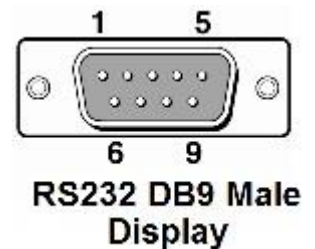
Stop Bits: One

Flow Control: None

Pin Assignments Diagram



Signal	Main Cable	Pin	Signal
TxD		2	TxD
RxD		3	RxD
GND		5	GND



Network communication

When configuring the Ethernet module, be sure device settings match those of the Global Scripter ethernet interface

Port Type:	Ethernet
Default Port:	9761
Logon Credentials Supported:	No
Multi-Connection Capabilities:	Undetermined
Port Changeability:	Undetermined

Ethernet Module Configuration Description

Please refer to user manual for settings and changes to the network communication

Notes for the Device

Appendix A. Set Commands (Serial)

Aspect Ratio 16:9 Device ID 1	kc 01 02\x0D
Aspect Ratio 16:9 Device ID 99	kc 63 02\x0D
Aspect Ratio 4:3 Device ID 1	kc 01 01\x0D
Aspect Ratio 4:3 Device ID 99	kc 63 01\x0D
Auto Image None Device ID 1	ju 01 01\x0D
Auto Image None Device ID 99	ju 63 01\x0D
Executive Mode Off Device ID 1	km 01 00\x0D
Executive Mode Off Device ID 99	km 63 00\x0D
Executive Mode On Device ID 1	km 01 01\x0D
Executive Mode On Device ID 99	km 63 01\x0D
Input AV Device ID 1	xb 01 20\x0D
Input AV Device ID 99	xb 63 20\x0D
Input Component Device ID 1	xb 01 40\x0D
Input Component Device ID 99	xb 63 40\x0D
Input DisplayPort Device ID 1	xb 01 D0\x0D
Input DisplayPort Device ID 99	xb 63 D0\x0D
Input DVI-D Device ID 1	xb 01 70\x0D
Input DVI-D Device ID 99	xb 63 70\x0D
Input HDMI Device ID 1	xb 01 A0\x0D
Input HDMI Device ID 99	xb 63 A0\x0D
Input RGB Device ID 1	xb 01 60\x0D
Input RGB Device ID 99	xb 63 60\x0D
Keypad 0 Device ID 1	mc 01 10\x0D
Keypad 0 Device ID 99	mc 63 10\x0D
Keypad 9 Device ID 1	mc 01 19\x0D
Keypad 9 Device ID 99	mc 63 19\x0D
Menu Navigation Back Device ID 1	mc 01 28\x0D
Menu Navigation Back Device ID 99	mc 63 28\x0D
Menu Navigation Down Device ID 1	mc 01 41\x0D
Menu Navigation Down Device ID 99	mc 63 41\x0D
Menu Navigation Enter Device ID 1	mc 01 44\x0D
Menu Navigation Enter Device ID 99	mc 63 44\x0D
Menu Navigation Exit Device ID 1	mc 01 5B\x0D
Menu Navigation Exit Device ID 99	mc 63 5B\x0D
Menu Navigation Left Device ID 1	mc 01 07\x0D
Menu Navigation Left Device ID 99	mc 63 07\x0D

**Global Scripter Module
Communication Sheet**

Revision: 10/5/2018

Menu Navigation Menu Device ID 1	mc 01 43\x0D
Menu Navigation Menu Device ID 99	mc 63 43\x0D
Menu Navigation Right Device ID 1	mc 01 06\x0D
Menu Navigation Right Device ID 99	mc 63 06\x0D
Menu Navigation Up Device ID 1	mc 01 40\x0D
Menu Navigation Up Device ID 99	mc 63 40\x0D
Power Off Device ID 1	ka 01 00\x0D
Power Off Device ID 99	ka 63 00\x0D
Power On Device ID 1	ka 01 01\x0D
Power On Device ID 99	ka 63 01\x0D
Tile Mode None Column 1 Row 1 Device ID 1	dd 01 11\x0D
Tile Mode None Column 1 Row 1 Device ID 99	dd 63 11\x0D
Tile Mode None Column 1 Row 15 Device ID 1	dd 01 1F\x0D
Tile Mode None Column 1 Row 15 Device ID 99	dd 63 1F\x0D
Tile Mode None Column 15 Row 1 Device ID 1	dd 01 F1\x0D
Tile Mode None Column 15 Row 1 Device ID 99	dd 63 F1\x0D
Tile Mode None Column 15 Row 15 Device ID 1	dd 01 FF\x0D
Tile Mode None Column 15 Row 15 Device ID 99	dd 63 FF\x0D
Tile Position None Tile ID 1 Device ID 1	di 01 01\x0D
Tile Position None Tile ID 1 Device ID 99	di 63 01\x0D
Tile Position None Tile ID 225 Device ID 1	di 01 E1\x0D
Tile Position None Tile ID 225 Device ID 99	di 63 E1\x0D

Appendix B. Update Commands (Serial)

Aspect Ratio Device ID 1	kc 01 FF\x0D
Aspect Ratio Device ID 99	kc 63 FF\x0D
Executive Mode Device ID 1	km 01 FF\x0D
Executive Mode Device ID 99	km 63 FF\x0D
Input Device ID 1	xb 01 FF\x0D
Input Device ID 99	xb 63 FF\x0D
Power Device ID 1	ka 01 FF\x0D
Power Device ID 99	ka 63 FF\x0D

Appendix C. Set Commands (Ethernet)

Aspect Ratio 16:9	kc 01 02\x0D
Aspect Ratio 4:3	kc 01 01\x0D
Auto Image None	ju 01 01\x0D
Executive Mode Off	km 01 00\x0D
Executive Mode On	km 01 01\x0D
Input AV	xb 01 20\x0D
Input Component	xb 01 40\x0D
Input DisplayPort	xb 01 D0\x0D
Input DVI-D	xb 01 70\x0D
Input HDMI	xb 01 A0\x0D
Input RGB	xb 01 60\x0D
Keypad 0	mc 01 10\x0D
Keypad 9	mc 01 19\x0D
Menu Navigation Back	mc 01 28\x0D
Menu Navigation Down	mc 01 41\x0D
Menu Navigation Enter	mc 01 44\x0D
Menu Navigation Exit	mc 01 5B\x0D
Menu Navigation Left	mc 01 07\x0D
Menu Navigation Menu	mc 01 43\x0D
Menu Navigation Right	mc 01 06\x0D
Menu Navigation Up	mc 01 40\x0D
Power Off None	ka 01 00\x0D
Tile Mode None Column 1 Row 1	dd 01 11\x0D
Tile Mode None Column 1 Row 15	dd 01 1F\x0D
Tile Mode None Column 15 Row 1	dd 01 F1\x0D
Tile Mode None Column 15 Row 15	dd 01 FF\x0D
Tile Position None Tile ID 1	di 01 01\x0D
Tile Position None Tile ID 225	di 01 E1\x0D